

# Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo + Discrepancy

Improving Efficiency

Selected Topics

Git website and repository  
Canvas

Fred Hickernell, Fall 2025

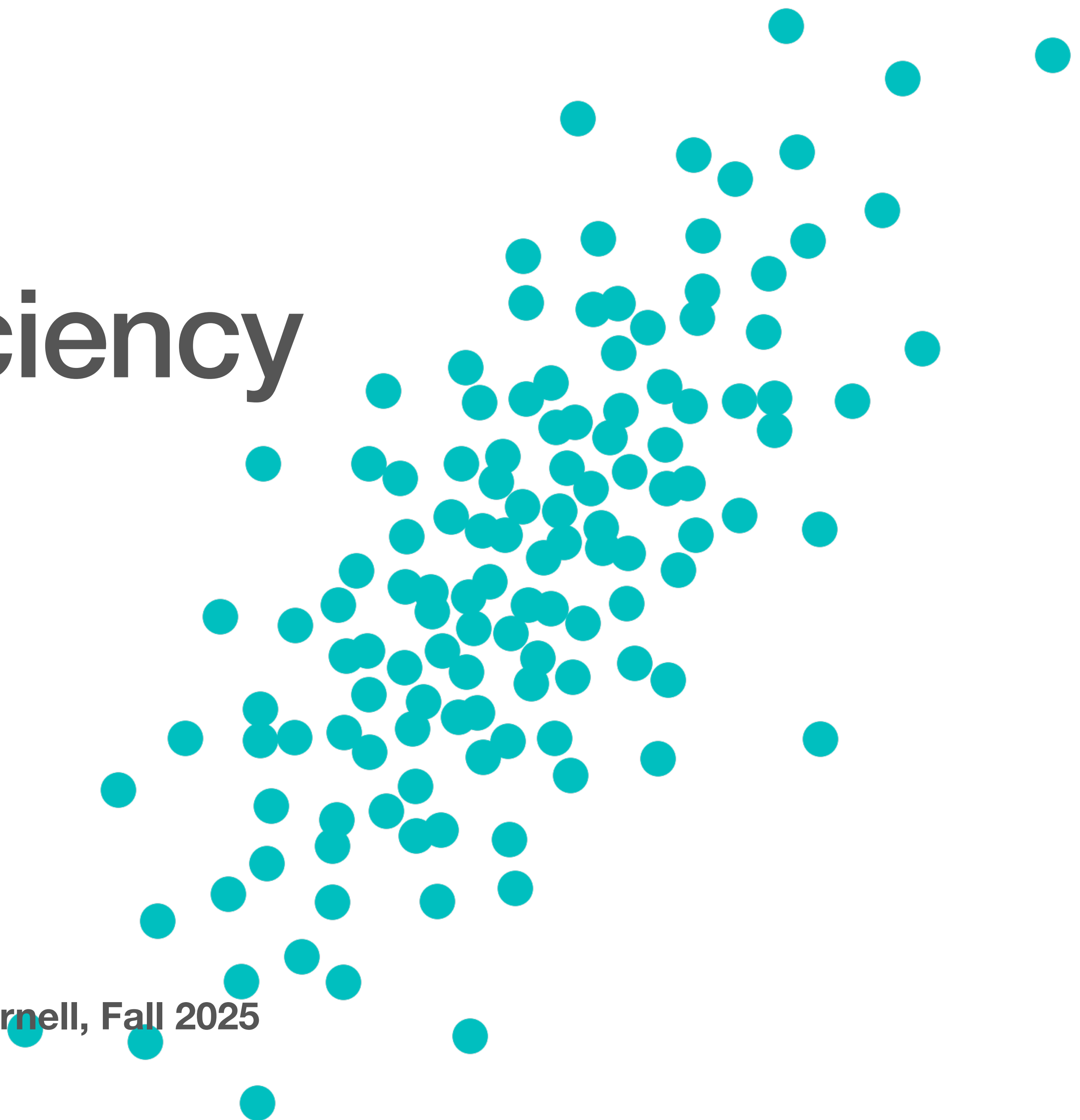
Updated 2025 November 13

# Improving Efficiency

Owen, Chapters 8–10, 15–17

Test 2 Oct 29

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025





# Variance Reduction

$$\mu = \mathbb{E}[f(X)] = \mathbb{E}[g(X)], \quad X \sim \mathcal{U}[0,1]^d$$

Is it better to use

$$\hat{\mu}_{f,n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \quad \text{or} \quad \hat{\mu}_{g,n} = \frac{1}{n} \sum_{i=0}^{n-1} g(X_i), \quad X_0, X_1, \dots \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}[0,1]^d$$

to approximate  $\mu$ ? Since both sample means are unbiased, we can compare their variances (mean squared errors):

$$\text{var}(\hat{\mu}_{f,n}) = \frac{\text{var}(f(X))}{n} \quad \text{var}(\hat{\mu}_{g,n}) = \frac{\text{var}(g(X))}{n}$$

so it depends on the variances of the two functions





# Keister example

$$\mu = \int_{\mathbb{R}^d} \cos(\|t\|) \exp(-\|t\|^2) dt$$

$$\mathbf{x} = (\Phi(at_1), \dots, \Phi(at_d)), \quad \Phi \text{ is the CDF of the standard normal}$$

$$d\mathbf{x} = a^d \phi(at_1) \cdots \phi(at_d) dt = \frac{a^d \exp(-a^2 \|t\|^2/2)}{(2\pi)^{d/2}} dt$$

$$t = (\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d)) / a$$

$$\begin{aligned} \mu = \int_{[0,1]^d} & \frac{(2\pi)^{d/2}}{a^d} \cos\left(\left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\| / a\right) \\ & \times \exp\left(-\frac{2-a^2}{2a^2} \left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\|^2\right) d\mathbf{x}, \quad (a^2 \leq 2) \end{aligned}$$

Whole family  $f(\mathbf{x}; a)$  for which  $\mu = \mathbb{E}[f(X; a)]$  for  $X \sim \mathcal{U}[0,1]^d$



# Variable transformations

Suppose that you want to approximate

$$\mu = \int_{\mathcal{T}} g(\mathbf{t}) \, d\mathbf{t}$$

by Monte Carlo. You need to write it as an expectation. Let  $\mathbf{t} = \boldsymbol{\psi}(\mathbf{x})$ , for some **invertible transformation**,  $\boldsymbol{\psi} : \mathcal{X} \rightarrow \mathcal{T}$ . Define the **Jacobian determinant** as

$$J(\mathbf{x}; \boldsymbol{\psi}) := \det \left( \left( \frac{\partial \psi_i}{\partial x_j} \right)_{i,j=1}^d \right), \quad J(\mathbf{t}; \boldsymbol{\psi}^{-1}) := \det \left( \left( \frac{\partial \psi_i^{-1}}{\partial t_j} \right)_{i,j=1}^d \right), \quad J(\boldsymbol{\psi}(\mathbf{x}); \boldsymbol{\psi}^{-1}) = \frac{1}{J(\mathbf{x}; \boldsymbol{\psi})}$$

Then

$$\mu = \int_{\mathcal{X}} g(\boldsymbol{\psi}(\mathbf{x})) |J(\mathbf{x}; \boldsymbol{\psi})| \, d\mathbf{x} = \int_{\mathcal{X}} \underbrace{\frac{g(\boldsymbol{\psi}(\mathbf{x}))}{\varrho_{\mathbf{X}}(\mathbf{x})} |J(\mathbf{x}; \boldsymbol{\psi})|}_{f(\mathbf{x})} \varrho_{\mathbf{X}}(\mathbf{x}) \, d\mathbf{x} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$



# Variable transformations

$$\mu = \int_{\mathcal{T}} g(t) dt = ?$$

Let  $t = \psi(x)$ , for some **invertible transformation**,  $\psi : \mathcal{X} \rightarrow \mathcal{T}$ . Then

$$\mu = \int_{\mathcal{X}} g(\psi(x)) |J(x; \psi)| dx = \int_{\mathcal{X}} \underbrace{\frac{g(\psi(x))}{\varrho(x)} |J(x; \psi)|}_{f(x)} \varrho(x) dx = \mathbb{E}_{\mathbf{X} \sim \varrho}[f(\mathbf{X})]$$

If  $\mathbf{X} \sim \mathcal{U}[0,1]^d$  (because we know how generate **uniform** samples), then

$$\mu = \int_{[0,1]^d} \underbrace{g(\psi(x)) |J(x; \psi)|}_{f(x)} dx = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d}[f(\mathbf{X})]$$





# Transformations of random variables

Let

- $T$  be a random variable with sample space  $\mathcal{T}$  and density  $\varrho_T$
- $X$  be a random variable with sample space  $\mathcal{X}$  and density  $\varrho_X$
- $T = \psi(X)$ , for some **invertible transformation**,  $\psi : \mathcal{X} \rightarrow \mathcal{T}$ . Then

$$\varrho_T(t) = \varrho_X(\psi^{-1}(t)) |J(t; \psi^{-1})|, \quad \varrho_T(\psi(x)) = \varrho_X(x) |J(\psi(x)); \psi^{-1})| = \frac{\varrho_X(x)}{|J(x; \psi)|}$$

$$\mu = \int_{\mathcal{T}} \underbrace{h(t) \varrho_T(t)}_{g(t)} dt = \mathbb{E}_{T \sim \varrho_T}[h(T)] = \mathbb{E}_{X \sim \varrho_X}[h(\psi(X))] = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$

$$\text{or } = \int_{\mathcal{X}} \frac{h(\psi(x)) \varrho_T(\psi(x))}{\varrho_X(x)} |J(x; \psi)| \varrho_X(x) dx = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$



# Importance sampling

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \int_{\mathcal{T}} \underbrace{h(\mathbf{t}) \frac{\varrho_{\mathbf{T}}(\mathbf{t})}{\varrho_{\mathbf{X}}(\mathbf{t})}}_{f(\mathbf{t})} \overbrace{\varrho_{\mathbf{X}}(\mathbf{t})}^{\text{likelihood ratio}} \, d\mathbf{t} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$

We try to choose  $\varrho_{\mathbf{X}}$  to make  $\text{var}[f(\mathbf{X})]$  small when doing IID sampling. If  $\mathbf{T} = \boldsymbol{\psi}(\mathbf{X})$  for  $\mathbf{X} \sim \mathcal{X}[0,1]^d$ —since often find generate non-uniform samples from uniform ones—then

$$\mu = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d} [h(\boldsymbol{\psi}(\mathbf{X}))] = \int_{[0,1]^d} h(\boldsymbol{\psi}(\mathbf{x})) \, d\mathbf{x}$$





# Importance sampling cont'd

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \int_{\mathcal{T}} \underbrace{h(\mathbf{t}) \frac{\varrho_{\mathbf{T}}(\mathbf{t})}{\varrho_{\mathbf{X}}(\mathbf{t})}}_{f(\mathbf{t})} \varrho_{\mathbf{X}}(\mathbf{t}) \, d\mathbf{t} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$

likelihood  
ratio

$$\hat{\mu}_{\text{IS},n} = \frac{1}{n} \sum_{i=0}^{n-1} h(X_i) \frac{\varrho_{\mathbf{T}}(X_i)}{\varrho_{\mathbf{X}}(X_i)}, \quad X_0, X_1, \dots \stackrel{\text{IID}}{\sim} \varrho_{\mathbf{X}}$$
$$\text{var}(\hat{\mu}_{\text{IS},n}) \approx \frac{1}{n(n-1)} \sum_{i=0}^{n-1} \left[ h(X_i) \frac{\varrho_{\mathbf{T}}(X_i)}{\varrho_{\mathbf{X}}(X_i)} - \hat{\mu}_{\text{IS},n} \right]^2$$



# Ex. Brownian motion with drift

In computational finance certain outcomes, such as positive option payoffs, may be **rare**

- Adding a drift to the Brownian motion **increases** the probability that asset paths yield positive payoffs and
- **Reduces the variance** of our estimator of the option price (population mean)
- Paths with positive payoffs receive **smaller sample weights** to compensate and keep the mean the same

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}, \quad \varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \left( \min(t_i, t_j) \right)_{i,j=1}^d$$

Here the density  $\varrho$  is a discretized **Brownian motion**



# Ex. Brownian motion with drift cont'd

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}$$
$$\varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \left( \min(t_i, t_j) \right)_{i,j=1}^d$$

We importance sample:

$$\varrho_{\text{drift}}(\mathbf{x}) = \frac{\exp(-(\mathbf{x} - \mathbf{a})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{a}) / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \mathbf{a} = \alpha(t_1, \dots, t_d)^\top$$

$$\text{likelihood ratio} = \frac{\varrho(\mathbf{x})}{\varrho_{\text{drift}}(\mathbf{x})} = \exp(-\alpha x_d + \alpha^2 t_d / 2)$$

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \exp(-\alpha x_d + \alpha^2 t_d / 2) \varrho_{\text{drift}}(\mathbf{x}) d\mathbf{x}$$

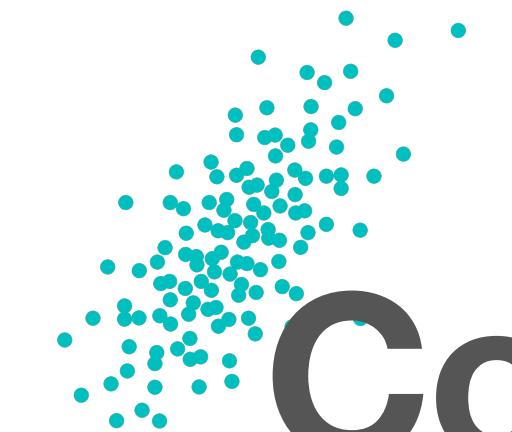
Note that the weight gets **smaller** as  $\alpha x_d$  increases or  $\alpha^2 t_d$  decreases





# Variable transformation / importance sampling observations

- Variable transformation and importance sampling are **intertwined**
- The **choice** of sampling density or variable transformation affects the variance of the sample mean
- You must choose the variable transformation so that
  - The Jacobian determinant,  $J(\mathbf{x}; \boldsymbol{\psi})$ , is **finite**
  - The probability density,  $q_{\mathbf{X}}(\mathbf{x})$ , does not **vanish** unless integrand vanishes
- You might try **pilot** samples to help choose transformation



# Control variates

$$\mu = \mathbb{E}[f(X)] \iff \mu = \mathbb{E}[f(X) - \beta_1\eta_1(X) - \dots - \beta_m\eta_m(X)]$$

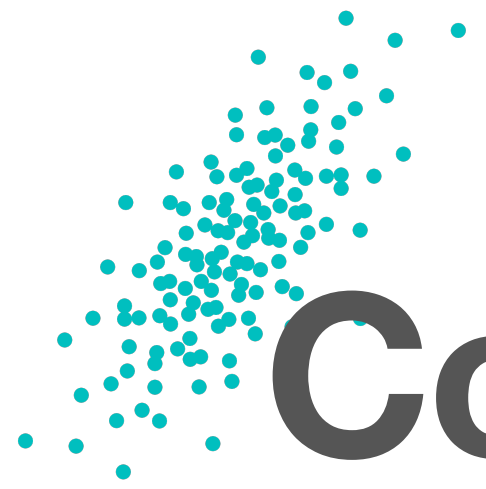
provided that  $\mathbb{E}[\eta_1(X)] = \dots \mathbb{E}[\eta_m(X)] = 0$ . Want to arrange such that

$$\text{var}[f(X) - \beta_1\eta_1(X) - \dots - \beta_m\eta_m(X)] \leq \text{var}[f(X)]$$

Then

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1\eta_1(X_i) - \dots - \beta_m\eta_m(X_i)] \text{ converges faster than } \hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i)$$
$$\text{var}(\hat{\mu}_{\text{CV},n}) \approx \frac{1}{n(n-1)} \sum_{i=0}^{n-1} [f(X_i) - \beta_1\eta_1(X_i) - \dots - \beta_m\eta_m(X_i) - \hat{\mu}_{\text{CV},n}]^2$$

How do we **minimize**  $\text{var}[f(X) - \beta_1\eta_1(X) - \dots - \beta_m\eta_m(X)]$ ?



# Control variates (cont'd)

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \cdots - \beta_m \eta_m(X_i)]$$

How do we **minimize**  $\text{var}[f(X) - \beta_1 \eta_1(X) - \cdots - \beta_m \eta_m(X)]$ ? Use least squares regression with centered response and explanatory variables to choose  $\hat{\beta}$ :

$$\hat{\beta} = \underset{\mathbf{b}}{\text{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2$$

$$\text{where } \mathbf{y} = \begin{pmatrix} f(\mathbf{x}_0) - \hat{\mu}_n \\ \vdots \\ f(\mathbf{x}_{n-1}) - \hat{\mu}_n \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} \eta_1(\mathbf{x}_0) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_0) - \hat{\mu}_{\eta_m,n} \\ \vdots & & \vdots \\ \eta_1(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_m,n} \end{pmatrix}$$

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i), \quad \hat{\mu}_{\eta_1,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_1(\mathbf{x}_i), \quad \dots, \quad \hat{\mu}_{\eta_m,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_m(\mathbf{x}_i)$$





# Least Squares

How do we solve

$$\hat{\beta} = \underset{\mathbf{b}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 = \underset{\mathbf{b}}{\operatorname{argmin}} (\mathbf{y} - \mathbf{H}\mathbf{b})^T (\mathbf{y} - \mathbf{H}\mathbf{b})$$

Let

$$\mathbf{b} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}$$

Then

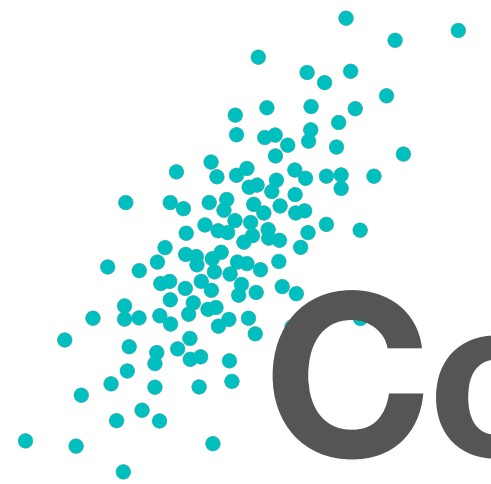
$$\begin{aligned} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 &= \|\mathbf{y} - \mathbf{H}[(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}]\|^2 \\ &= \|[ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{H}\mathbf{c}\|^2 \\ &= ([ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{H}\mathbf{c})^T ([ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{H}\mathbf{c}) \\ &= \mathbf{y}^T [ \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T ] \mathbf{y} + \mathbf{c}^T \mathbf{H}^T \mathbf{H} \mathbf{c} \end{aligned}$$

So the optimum comes by setting  $\mathbf{c} = \mathbf{0}$



# Control variates

- Should be **clever** in choosing the  $\eta_j$ 
  - **Good** choices can decrease the sample size required
  - **Bad** choices only hurt in terms of increased **computation** time, but not increased sample size
- The  $\beta_j$  should be chosen by **least squares** for IID sampling
- For **low discrepancy** sequences, the  $\beta_j$  should be chosen to reduce the **variation** (not variance)



# Conditional Monte Carlo

Recall that

$$\mathbb{E}(Y) = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})], \quad \text{var}(Y) = \mathbb{E}_{\mathbf{Z}}[\text{var}(Y | \mathbf{Z})] + \text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})]$$

Which implies that

$$\text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})] \leq \text{var}(Y), \quad \text{var} \left( \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_Y(Y | \mathbf{Z}_i) \right) \leq \text{var} \left( \frac{1}{n} \sum_{i=0}^{n-1} Y_i \right)$$

Let  $(Y = f(\mathbf{X}_{1:d}), \mathbf{X}_{1:d})$  be a vector of random variables

Let  $\mathbf{Z} = \mathbf{X}_{2:d}$

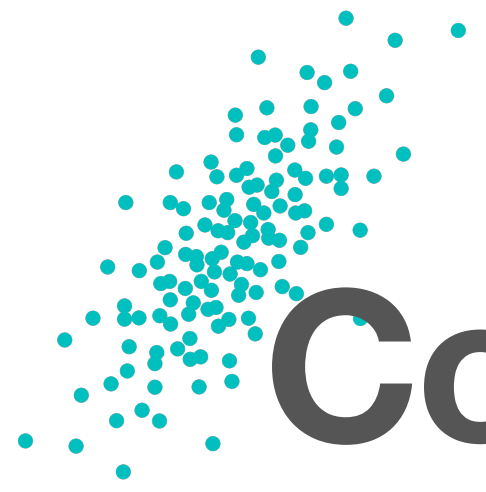
If  $g(\mathbf{X}_{2:d}) = \mathbb{E}_Y(Y | \mathbf{X}_{2:d})$  has an analytic form, then

$$\mu = \mathbb{E}(Y) = \mathbb{E}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})] \text{ and } \text{var}(Y) \geq \text{var}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})]$$

If  $h(y, \mathbf{X}_{2:d}) = \varrho_{Y|\mathbf{X}_{2:d}}(y | \mathbf{x}_{2:d})$  has an analytic form, then

$$\varrho_Y(y) = \mathbb{E}_{\mathbf{X}_{2:d}}[h(y, \mathbf{X}_{2:d})] \quad \text{probability density expressed as expectation}$$





# Conditional Monte Carlo for Density Estimation

Estimate the probability density,  $\varrho$ , for  $Y = f(X)$ , where  $X \sim \mathcal{U}[0,1]^d$

If one can identify an **increasing**  $g$ , such that

$$y = f(\mathbf{x}) \iff x_1 = g(y; \mathbf{x}_{2:d})$$

Then

$$\begin{aligned} F_{Y|X_{2:d}}(y | \mathbf{x}_{2:d}) &= \mathbb{P}[Y \leq y | X_{2:d} = \mathbf{x}_{2:d}] = \mathbb{P}[f(X) \leq y | X_{2:d} = \mathbf{x}_{2:d}] \\ &= \mathbb{P}[X_1 \leq g(y; \mathbf{x}_{2:d})] = g(y; \mathbf{x}_{2:d}) \end{aligned}$$

$$\varrho(y) = \mathbb{E}[\varrho_{Y|X_{2:d}}(y | \mathbf{x}_{2:d})] = \int_{[0,1]^{d-1}} g'(y; \mathbf{x}) \, d\mathbf{x}$$



## Ex. Density of a sum of uniforms

Let  $Y = \gamma_1 X_1 + \cdots + \gamma_d X_d$ , where  $X \sim \mathcal{U}[0,1]^d$  and all the  $\gamma_\ell$  are non-negative.

What is the **PDF** of  $Y$ ?

$$g(y; x_2, \dots, x_d) = \frac{1}{\gamma_1} (y - \gamma_2 x_2 - \cdots - \gamma_d x_d), \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \leq y \leq \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$

$$g'(y; x_2, \dots, x_d) = \frac{1}{\gamma_1}, \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \leq y \leq \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$



# Antithetic Sampling

If you are sampling from a probability distribution with symmetry, then you can evaluate  $f$  at both  $X_i$  and  $T(X_i)$ , where  $T(X)$  has the same distribution as  $X$ , e.g.,

$$T(X) = \mathbf{1} - X \text{ for } X \sim \mathcal{U}[0,1]^d \text{ or } T(X) = -X \text{ for } X \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

Note that if  $X_i$  are IID, then  $X_i$  and  $T(X_i)$  are not independent, but  $(X_i, T(X_i))$  are

When does  $\hat{\mu}_{\text{Anti},n} = \frac{1}{2n} \sum_{i=0}^{n-1} [f(X_i) + f(T(X_i))]$  have a smaller variance than

$$\hat{\mu}_{2n} = \frac{1}{2n} \sum_{i=0}^{2n-1} f(X_i) \text{ for IID } X_i?$$





# Antithetic Sampling

Note that

$$\text{var}(\hat{\mu}_{\text{Anti},n}) = \frac{1}{4n^2} \sum_{i=0}^{n-1} \text{var}(f(X_i) + f(T(X_i))) = \frac{\text{var}(f(X))}{2n} [1 + \text{corr}(f(X), f(T(X)))]$$

Whereas

$$\text{var}(\hat{\mu}_{2n}) = \frac{\text{var}(f(X))}{2n}$$

So antithetic sampling helps if  $\text{corr}(f(X), f(T(X))) < 0$

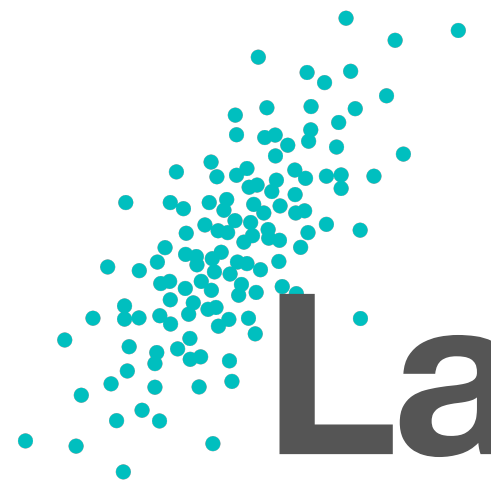
$$\text{var}(\hat{\mu}_{\text{Anti},n}) \approx \frac{1}{4n(n-1)} \sum_{i=0}^{n-1} [f(X_i) + f(T(X_i)) - 2\hat{\mu}_{\text{Anti},n}]^2$$



# How to get an antithetic variate

Suppose that you have random variable  $X$  with CDF  $F$ . One way to construct an **antithetic variate**,  $T(X)$ :

- Note that  $U = F(X) \sim \mathcal{U}[0,1]$
- Also,  $1 - U \sim \mathcal{U}[0,1]$
- Therefore  $T(X) = F^{-1}(1 - U) = F^{-1}(1 - F(X))$  also has CDF  $F$



# Latin Hypercube Sampling in One Dimension

Let's be intentional about placing uniform looking points on  $[0,1]$

$$X_i = (i + U_i)/n, \quad U_0, U_1, \dots \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1], \quad i = 0, \dots, n-1, \quad \hat{\mu}_{\text{LHS},n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i)$$

Note that  $X_i = (i + U_i)/n \sim \mathcal{U}[i/n, (i+1)/n]$ , and so

$$\bullet \quad \mathbb{E} [\hat{\mu}_{\text{LHS},n}] = \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[f(X_i)] = \frac{1}{n} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} f(x) \times n \times dx = \int_0^1 f(x) dx = \mu$$

which means that  $\hat{\mu}_{\text{LHS},n}$  is unbiased

- The  $X_i$  are not IID but they are independent





# Variance of LHS in One Dimension

$$\text{var}(\hat{\mu}_{\text{LHS},n}) = \text{var}\left(\sum_{i=0}^{n-1} \frac{f(X_i)}{n}\right) = \frac{1}{n^2} \sum_{i=0}^{n-1} \text{var}(f(X_i)) \quad \text{since the } X_i \text{ are uncorrelated}$$

$$= \frac{1}{n^2} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} (f(x) - f(\eta_i))^2 \times n \times dx$$

where  $f(\eta_i) = \mathbb{E}[f(X_i)]$  for some  $\eta_i$  by the Mean Value Theorem

$$= \frac{1}{n} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} (f'(\xi_i(x))(x - \eta_i))^2 dx,$$

where  $f(x) = f(\eta_i) + f'(\xi_i(x))(x - \eta_i)$  for some  $\xi(x)$  by Taylor's Theorem

$$\leq \frac{1}{n} \sum_{i=0}^{n-1} \frac{\|f'\|_{\infty}^2}{n^3} = \frac{\|f'\|_{\infty}^2}{n^3}$$

$$\text{rmse}(\hat{\mu}_{\text{LHS},n}) \leq \frac{\|f'\|_{\infty}}{n^{3/2}}$$



# Latin Hypercube Sampling in $d$ dimensions

$X_{i,\ell} = (\pi_\ell(i) + U_{i,\ell})/n$ ,  $U_{i,\ell} \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1]$ ,  $\pi_\ell$  are IID random permutations of  $1, \dots, n$

$$\hat{\mu}_{\text{LHS},n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_{i,1}, \dots, X_{i,d})$$

To quantify uncertainty in the estimate of the mean we need **replications**:

$X_{i,\ell}^{(r)} = (\pi_\ell^{(r)}(i) + U_{i,\ell}^{(r)})/n$ ,  $U_{i,\ell}^{(r)} \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1]$ ,  $\pi_\ell^{(r)} \stackrel{\text{IID}}{\sim}$ ,  $\hat{\mu}_{\text{LHS},n}^{(r)} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_{i,1}^{(r)}, \dots, X_{i,d}^{(r)})$ ,  $r = 1, \dots, R$

$$\hat{\mu}_{\text{LHS},n,R} = \frac{1}{R} \sum_{r=0}^{R-1} \hat{\mu}_{\text{LHS},n}^{(r)} = \frac{1}{nR} \sum_{r,i=0}^{R-1,n-1} f(X_{i,1}^{(r)}, \dots, X_{i,d}^{(r)})$$

$$\text{var}(\hat{\mu}_{\text{LHS},n,R}) \approx \frac{1}{R(R-1)} \sum_{r=0}^{R-1} \left[ \hat{\mu}_{\text{LHS},n}^{(r)} - \hat{\mu}_{\text{LHS},n,R} \right]^2$$



# Pros and cons of LHS

## Pros

- Do not need to **guess** importance sampling distributions or control variates or compute conditional expectations
- Can give **faster** convergence rates than IID

## Cons

- Sample size,  $n$ , is **fixed**
- Difficult to assess accuracy, except by **replications**; too many replications destroy the faster convergence
- Great convergence rates **do not hold** for multivariate case ( $d > 1$ )





# Low discrepancy sampling

Low discrepancy sampling, also known as [quasi-Monte Carlo methods](#) is an improvement on LHS that can provide [higher order convergence rates](#) than IID sampling for functions with modest smoothness

Recall from earlier the discrepancy,  $D(\{\mathbf{x}_i\}_{i=0}^{n-1}, F; K)$ , between a point cloud or node set  $\{\mathbf{x}_i\}_{i=0}^{n-1}$  and a probability distribution,  $F$ , defined by

$$D^2(\{\mathbf{x}_i\}_{i=0}^{n-1}, F; K) = \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF(\mathbf{x}) dF(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}_i) dF(\mathbf{x}) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{x}_i, \mathbf{x}_j),$$

Where we will take the domain to be  $\mathcal{X} = [0, 1]^d$



# Discrepancy interpretation

- Worst-case error

$$\left| \int_{[0,1]^d} f(\mathbf{x}) \, d(\mathbf{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i) \right| \leq D(\mathcal{U}[0,1]^d, \{\mathbf{x}_i\}_{i=0}^{n-1}; K) \|f\|_{\mathcal{H}} \quad \forall f \in \mathcal{H}$$

- Average-case error

$$\mathbb{E}_{f \in \mathcal{GP}(0,K)} \left| \int_{[0,1]^d} f(\mathbf{x}) \, d\mathbf{x} - \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i) \right|^2 = D^2(\mathcal{U}[0,1]^d, \{\mathbf{x}_i\}_{i=0}^{n-1}; K)$$

- Distance between the target distribution and the point cloud

$$\begin{aligned} \mathbb{E}_{\mathbf{X}, \mathbf{X}' \sim \mathcal{U}[0,1]^d} K(\mathbf{X}, \mathbf{X}') - 2 \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d, \mathbf{Z} \sim F_{\{\mathbf{x}_i\}_{i=0}^{n-1}}} K(\mathbf{X}, \mathbf{Z}) + \mathbb{E}_{\mathbf{Z}, \mathbf{Z}' \sim F_{\{\mathbf{x}_i\}_{i=0}^{n-1}}} K(\mathbf{Z}, \mathbf{Z}') \\ = D^2(\mathcal{U}[0,1]^d, \{\mathbf{x}_i\}_{i=0}^{n-1}; K) \end{aligned}$$



# van der Corput sequence

In one-dimension, evenly spaced points may be constructed as

$$x_i = i/n, \quad i = 0, \dots, n - 1$$

But you need to **fix  $n$**  in advance. The **van der Corput sequence** is an infinite sequence that fills the unit interval  $[0, 1]$  evenly and one can stop at any  $n$  and even keep going. Let  $b$  (normally  $b = 2$ ) be a prime base. Then

$$\phi_b(i) = \phi_b(\dots + i_2 b^2 + i_1 b + i_0) = \phi_b((\dots i_2 i_1 i_0)_b) := {}_b(0.i_0 i_1 i_2 \dots) = \frac{i_0}{b} + \frac{i_1}{b^2} + \frac{i_2}{b^3} + \dots$$

The set  $\{i/b^m\}_{i=0}^{b^m-1}$  and the set  $\{\phi_b(i)\}_{i=0}^{b^m-1}$  have the **same points** but in a different order, but the elements of the second set do not change their order when  $m$  increases



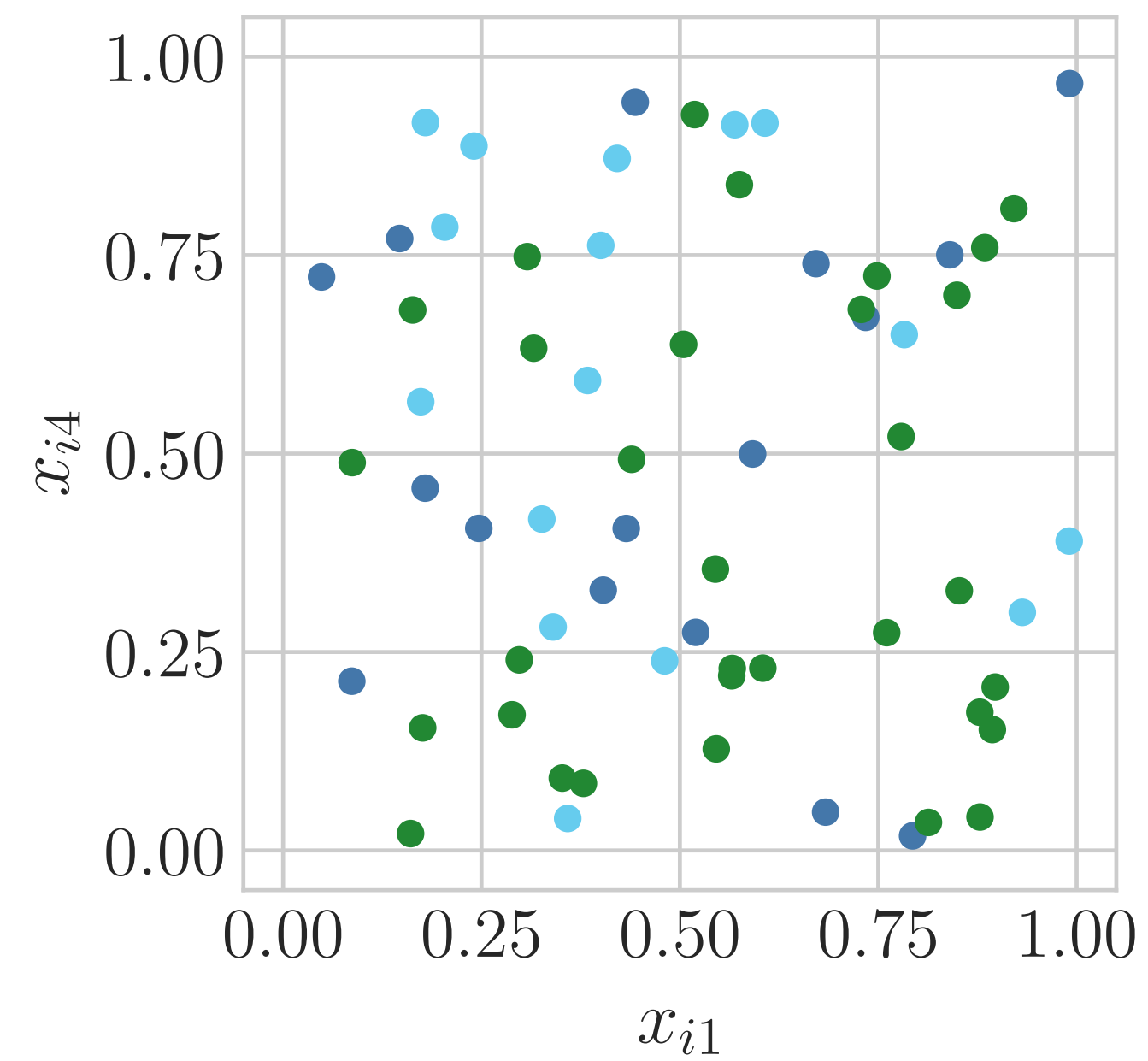
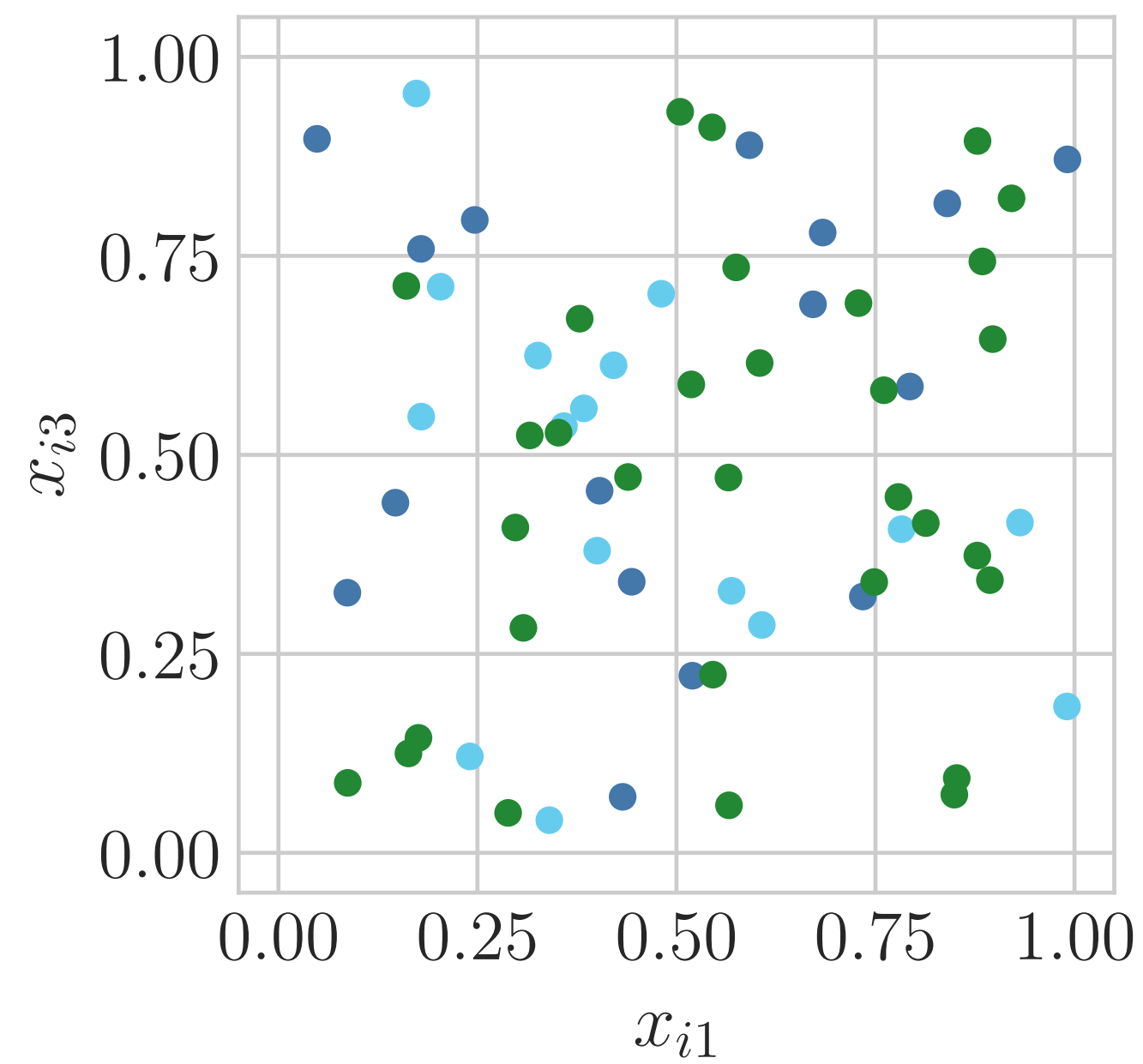
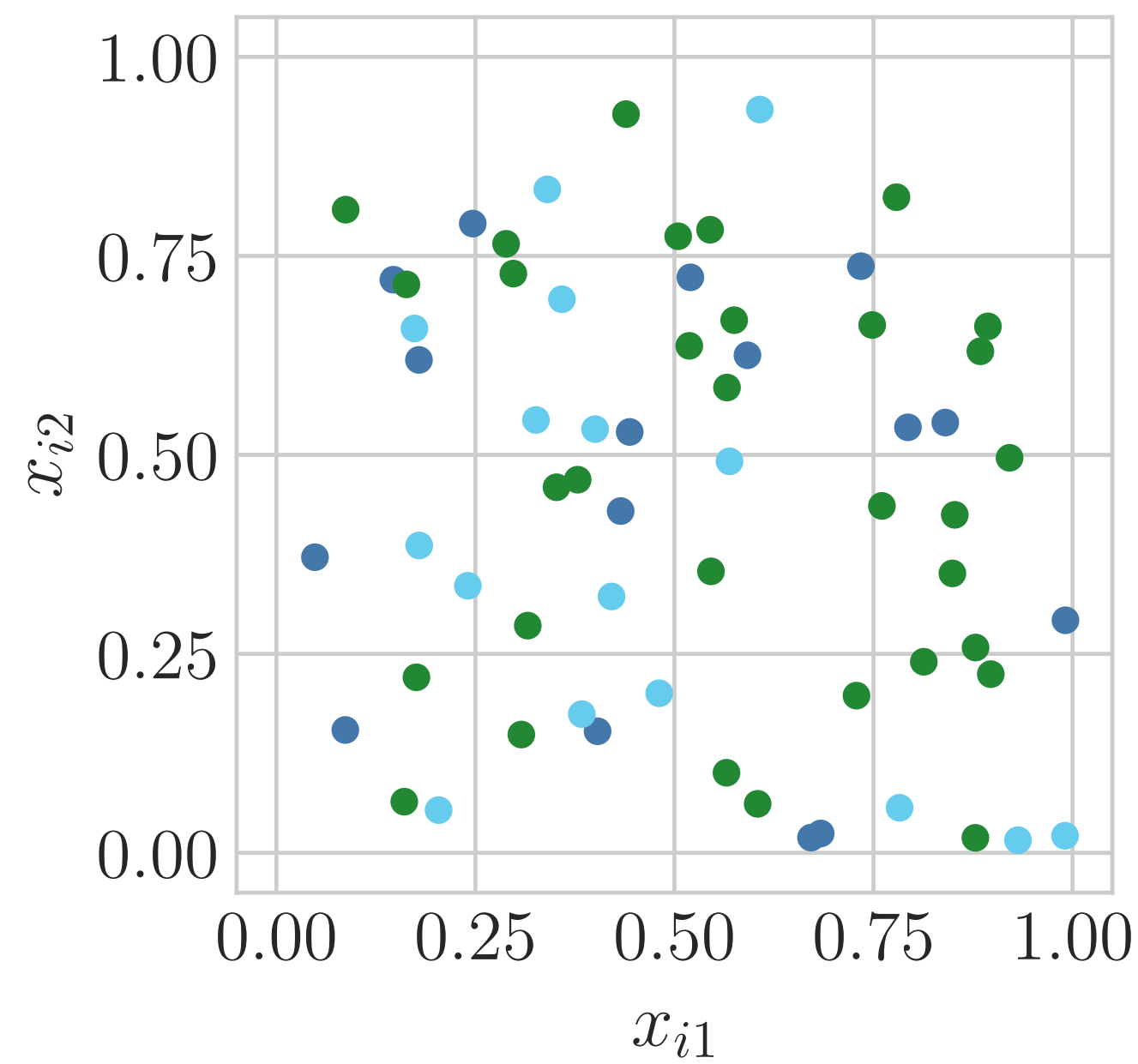


# Four families of low discrepancy sequences

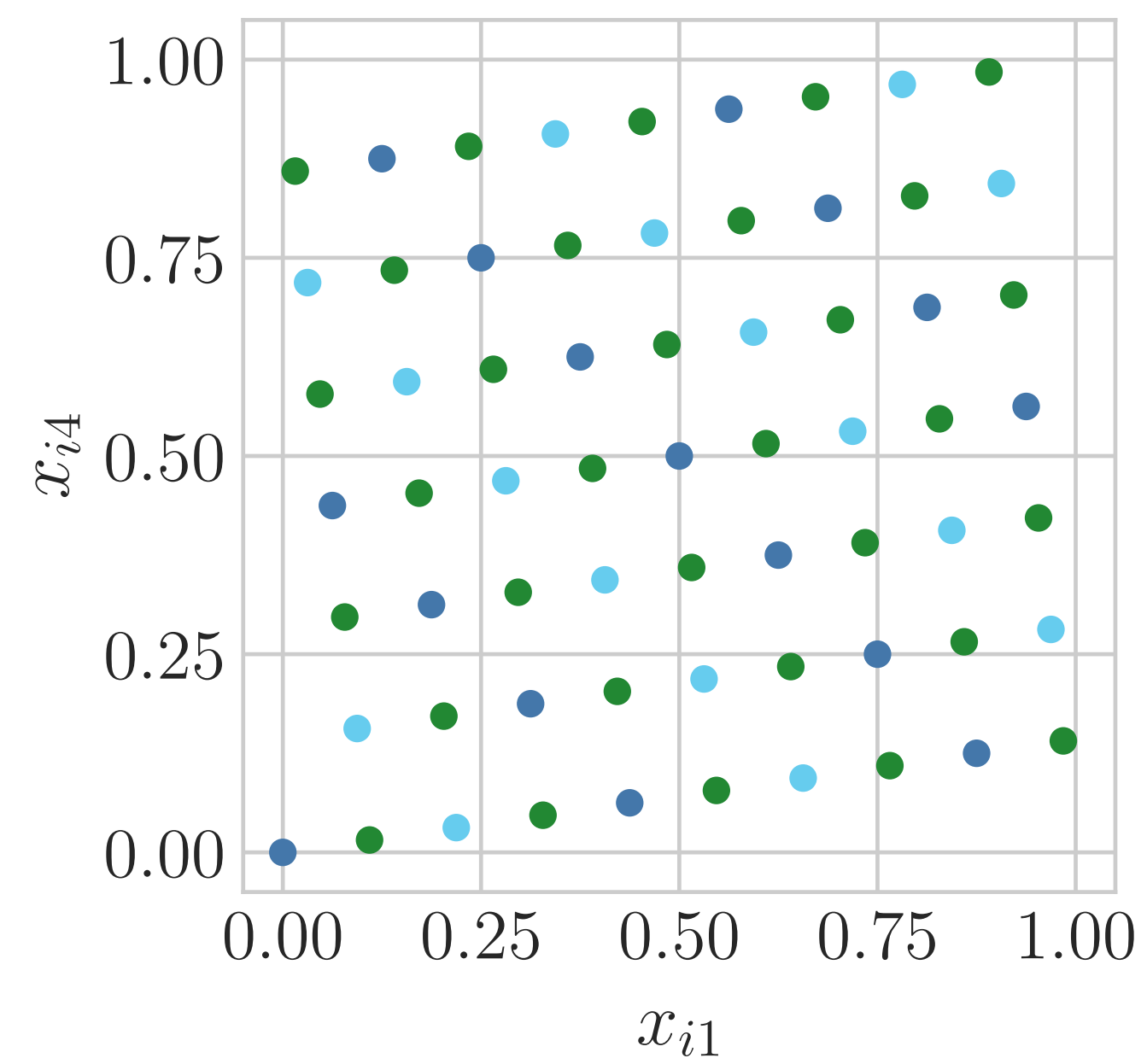
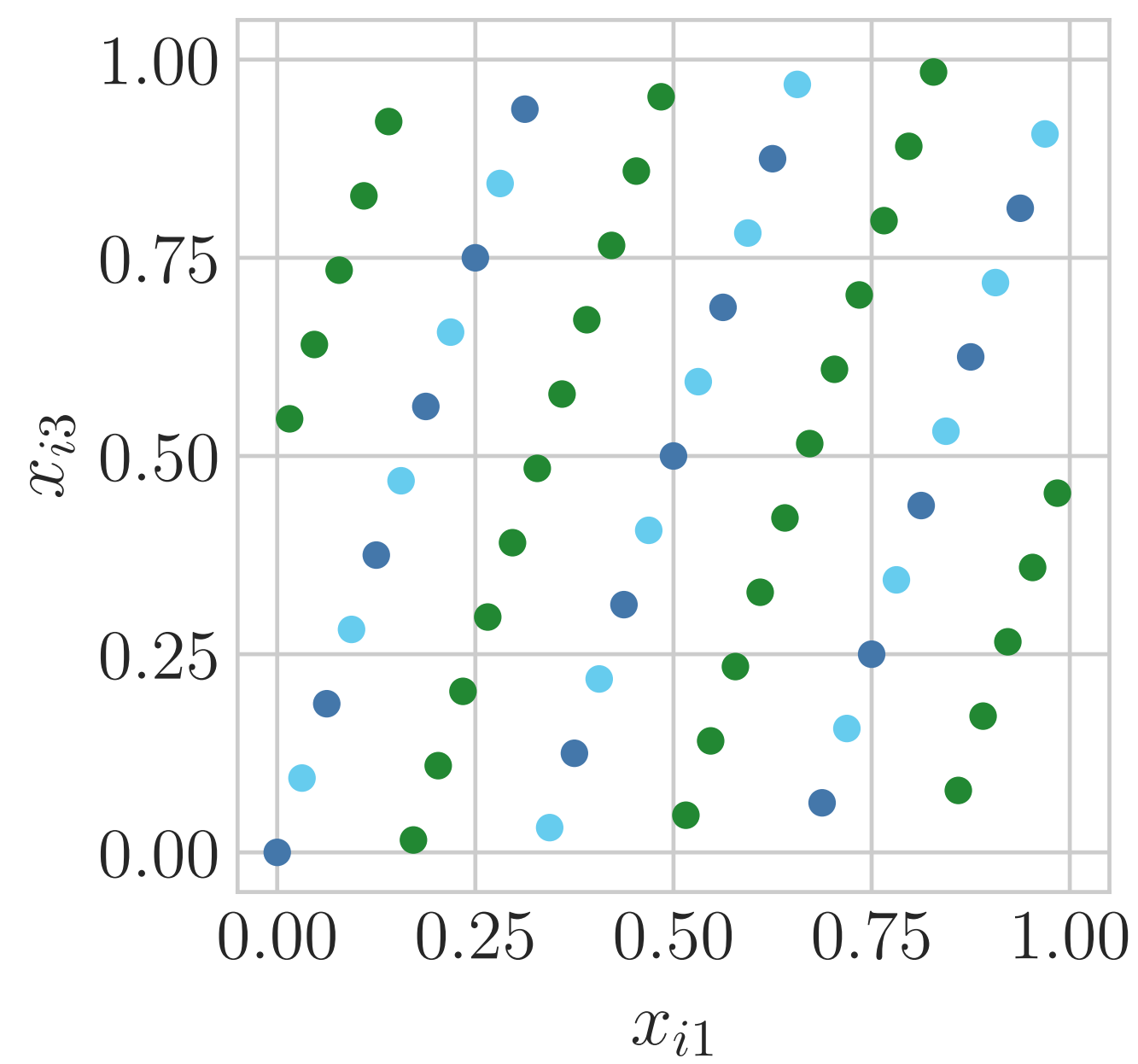
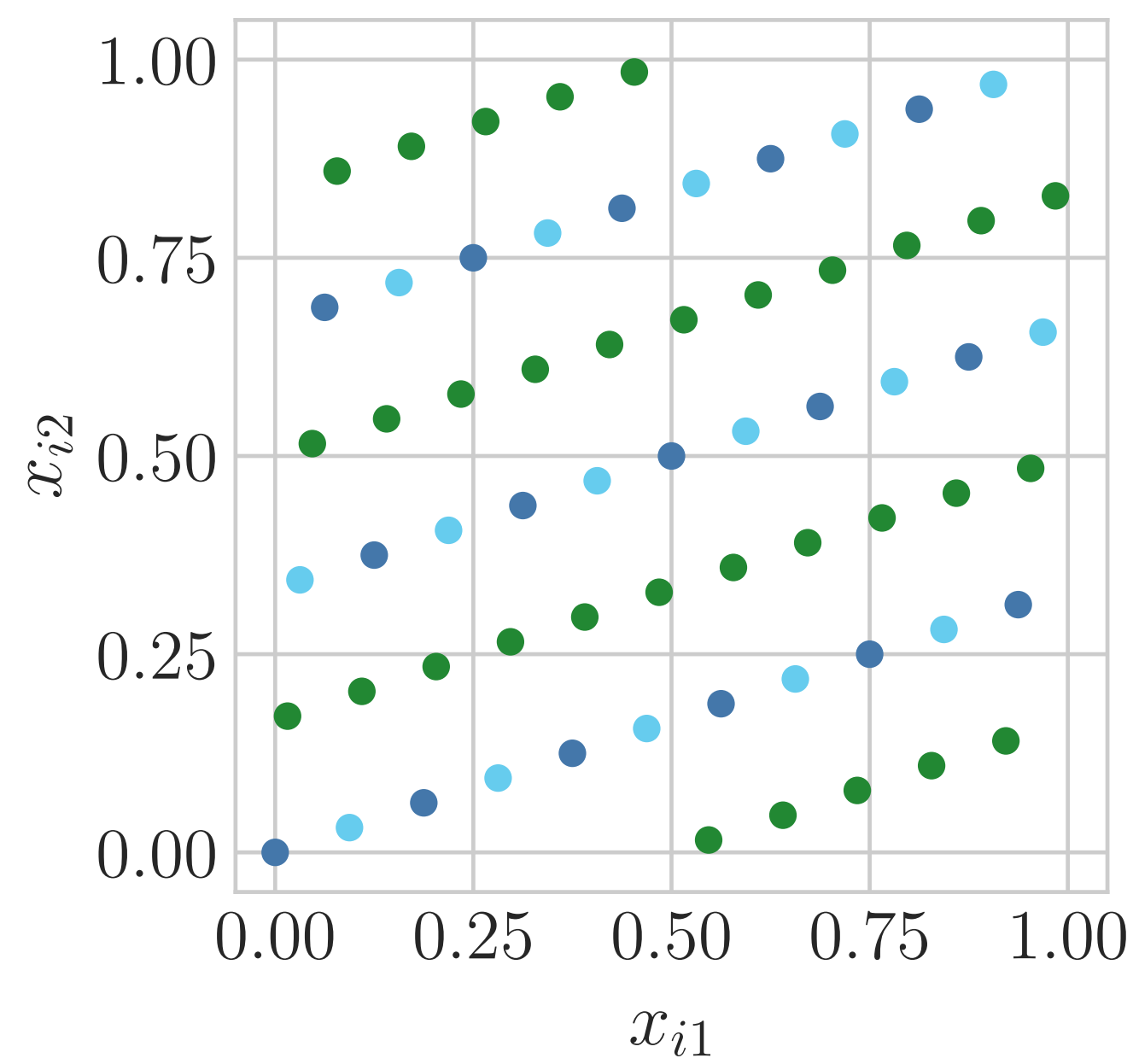
For  $i = 0, 1, \dots$

- **Kronecker**:  $x_i = i\zeta \bmod 1$  for some specially chosen generator  $\zeta \in [0, 1)^d$ ,  
no preferred sample size,  $n$
- **Lattice**:  $x_i = \phi_b(i)\zeta \bmod 1$  for some specially chosen generator  $\zeta \in \mathbb{N}^d$ ,  
preferred sample size is  $n = b^m$
- **Halton**:  $x_i = (\phi_2(i), \phi_3(i), \phi_5(i), \dots)$ , essentially no preferred sample size,  $n$
- **Digital**:  $x_i = i_0 c_0 \oplus i_1 c_1 \oplus i_2 c_2 \oplus \dots$ , where  $i = (\dots i_2 i_1 i_0)_b$ ,  $\oplus$  means  
digitwise addition modulo  $b$ , and  $c_0, c_1, c_2, \dots \in [0, 1)^d$ , preferred sample size  
is  $n = b^m$

IID

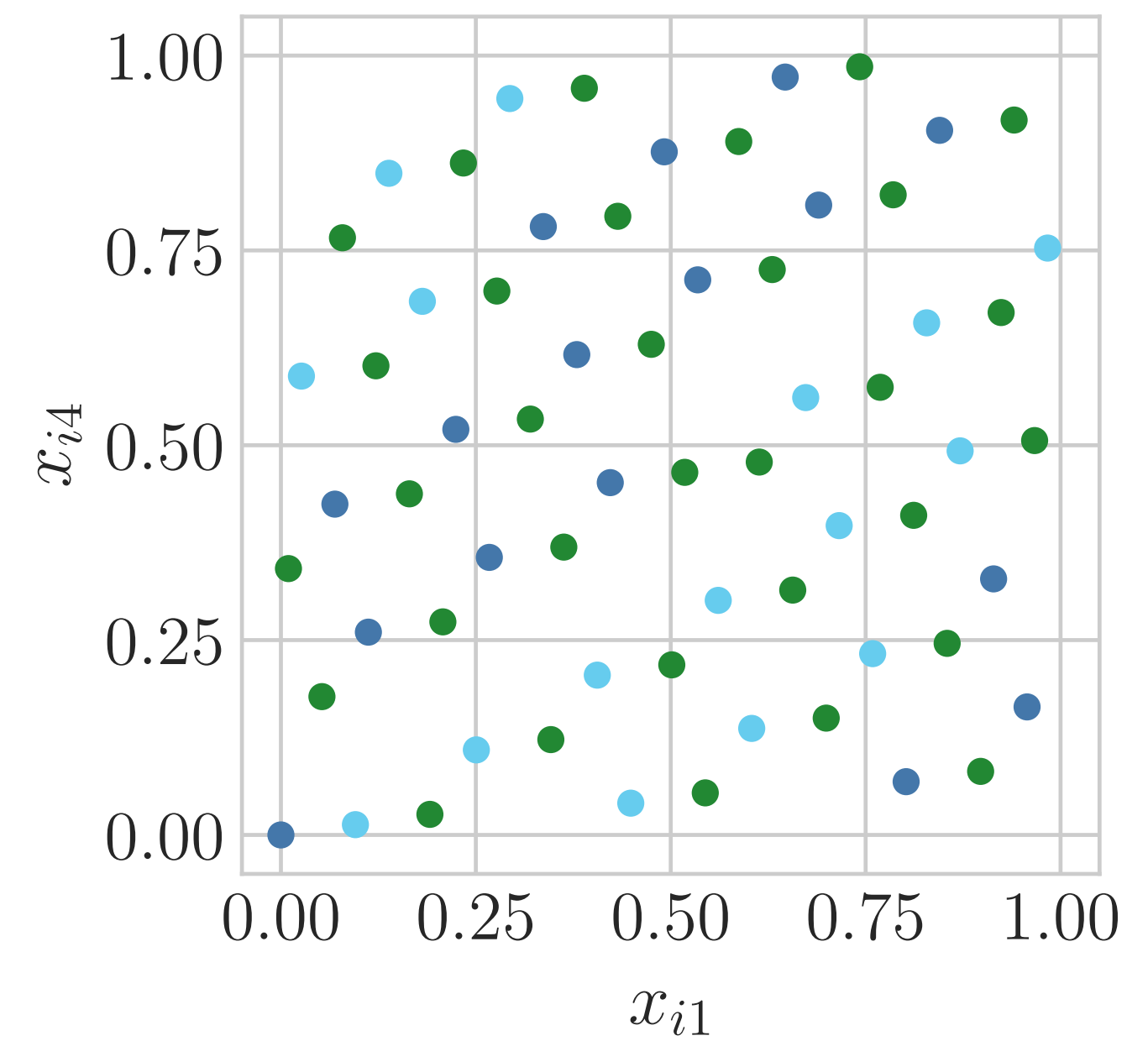
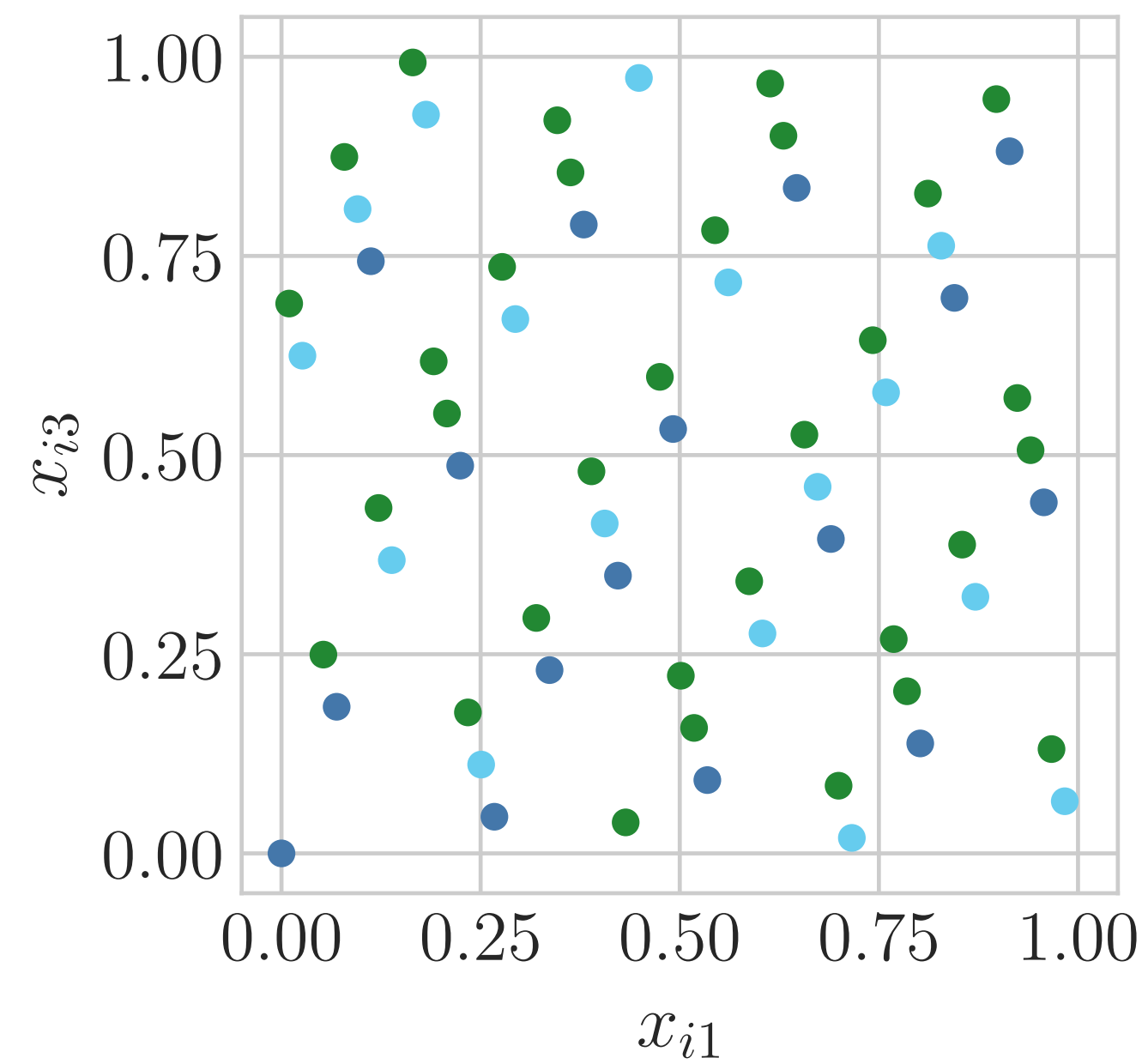
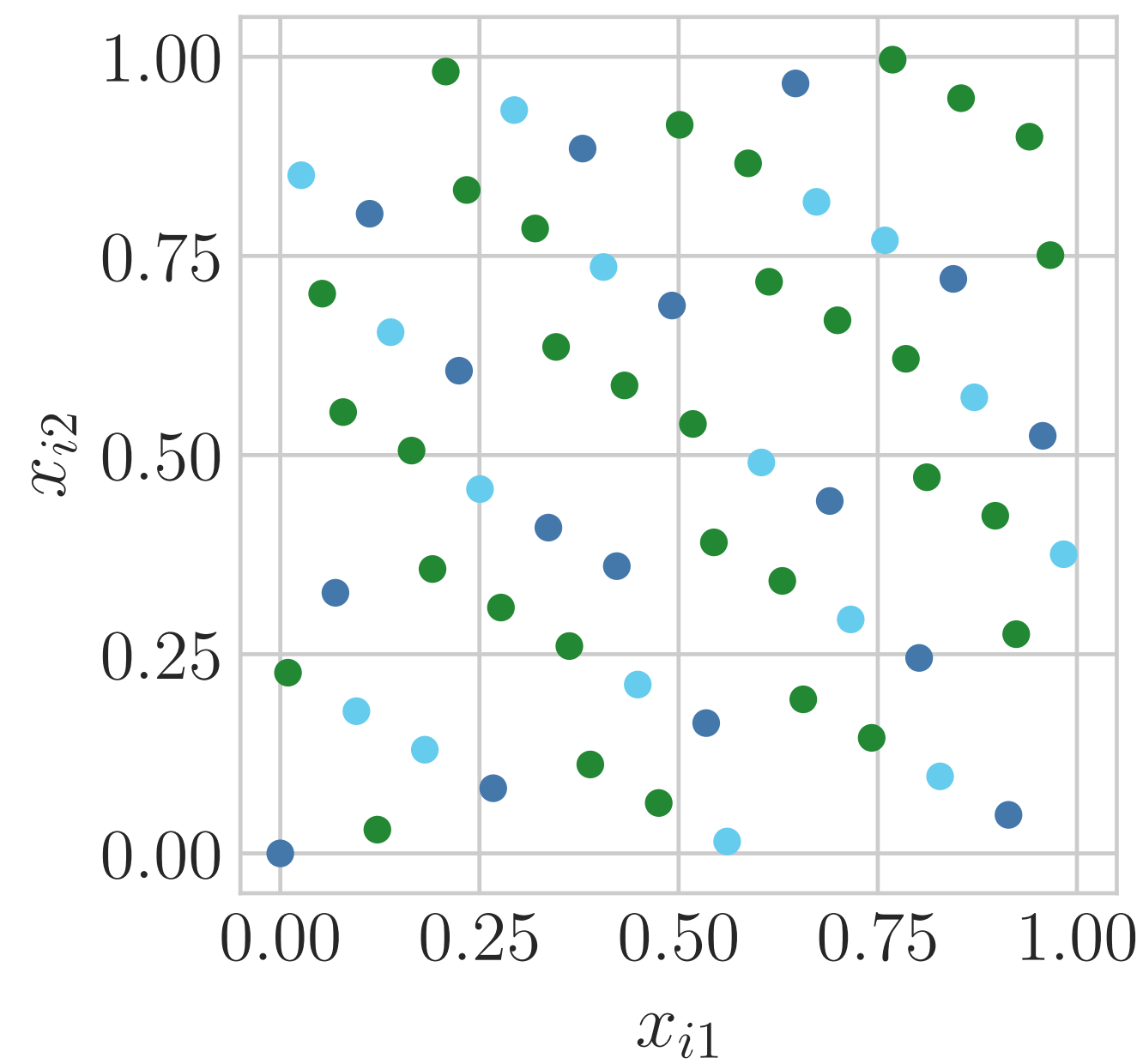


Lattice LD

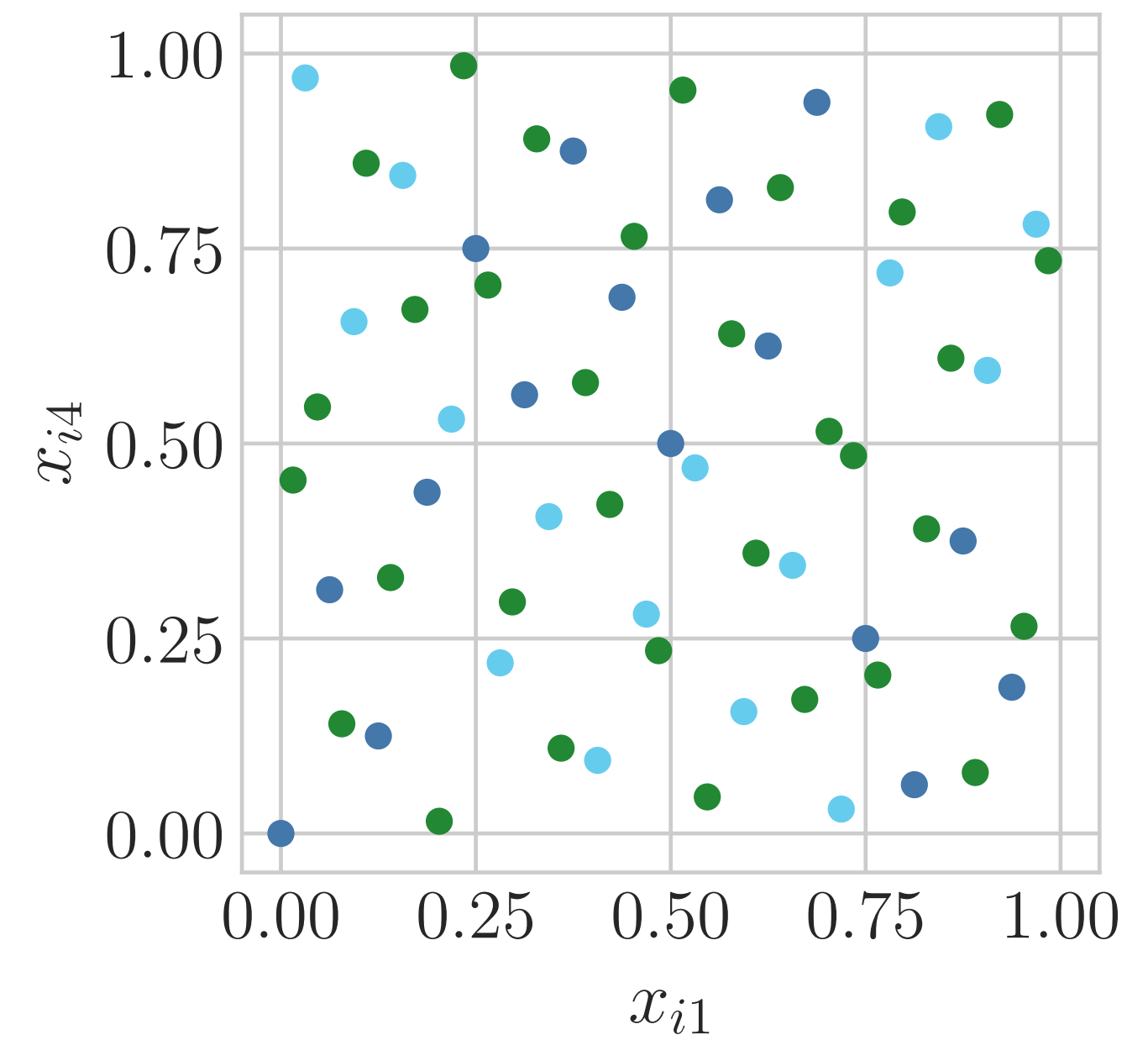
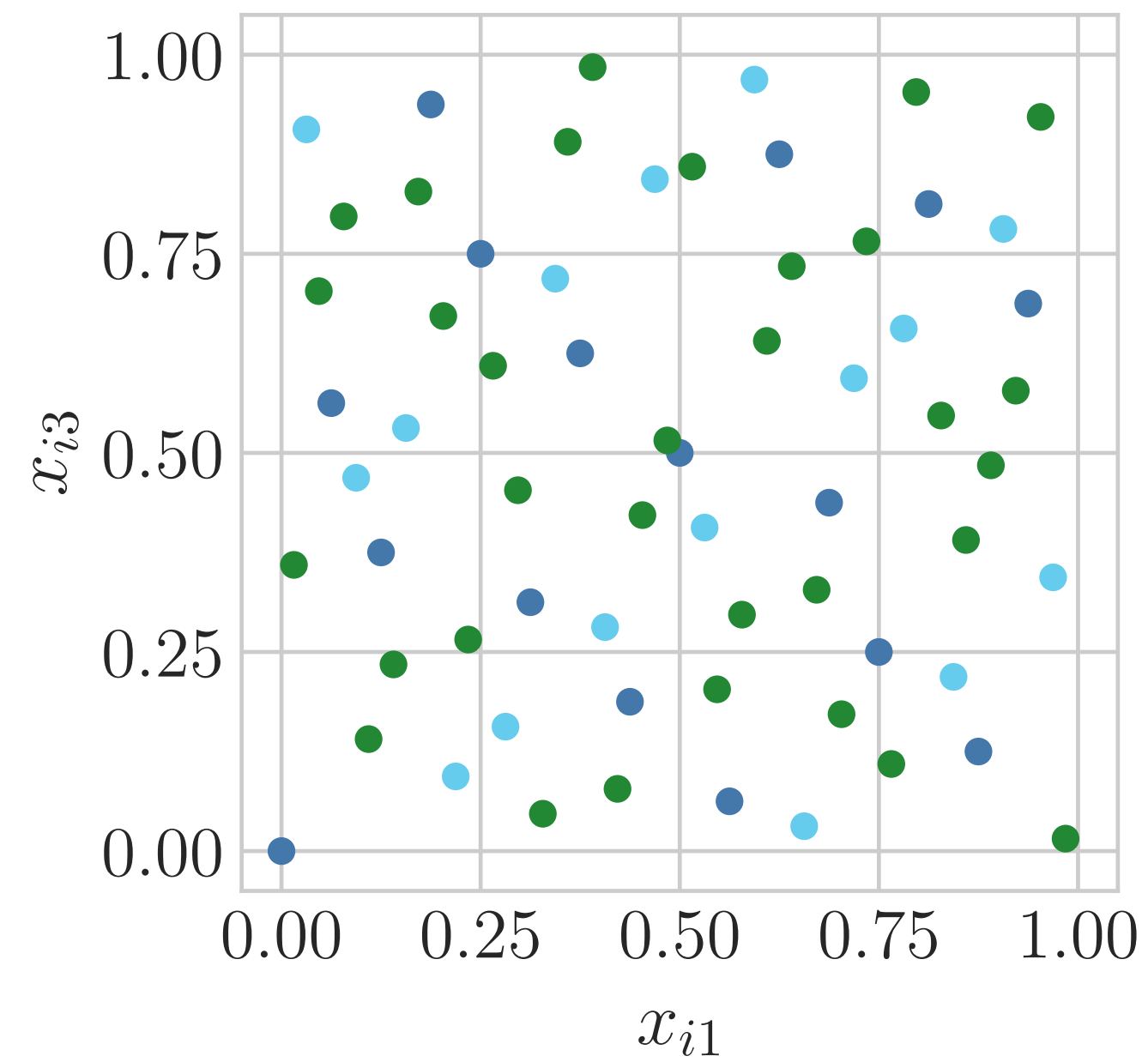
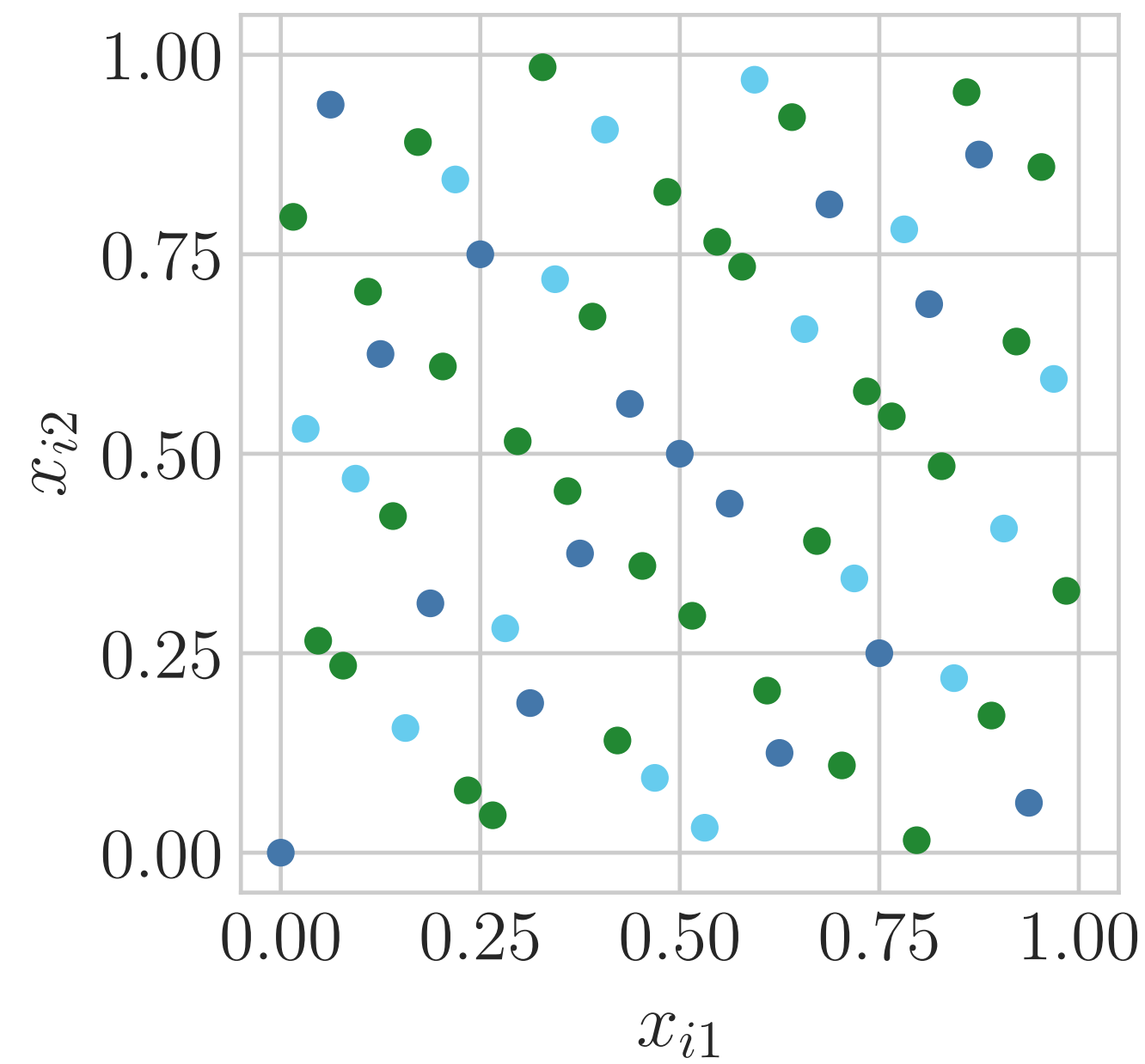




## Kronecker LD

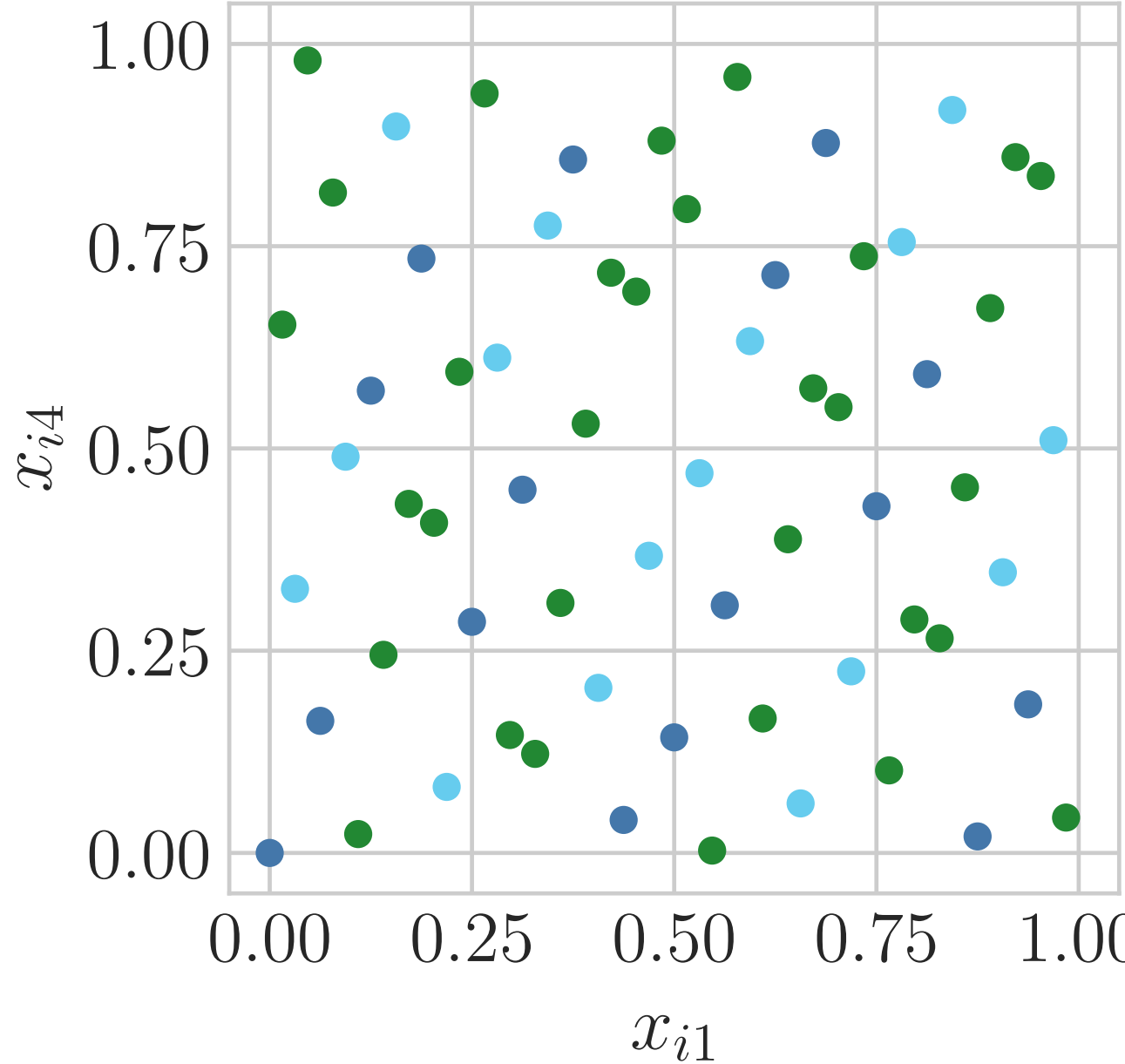
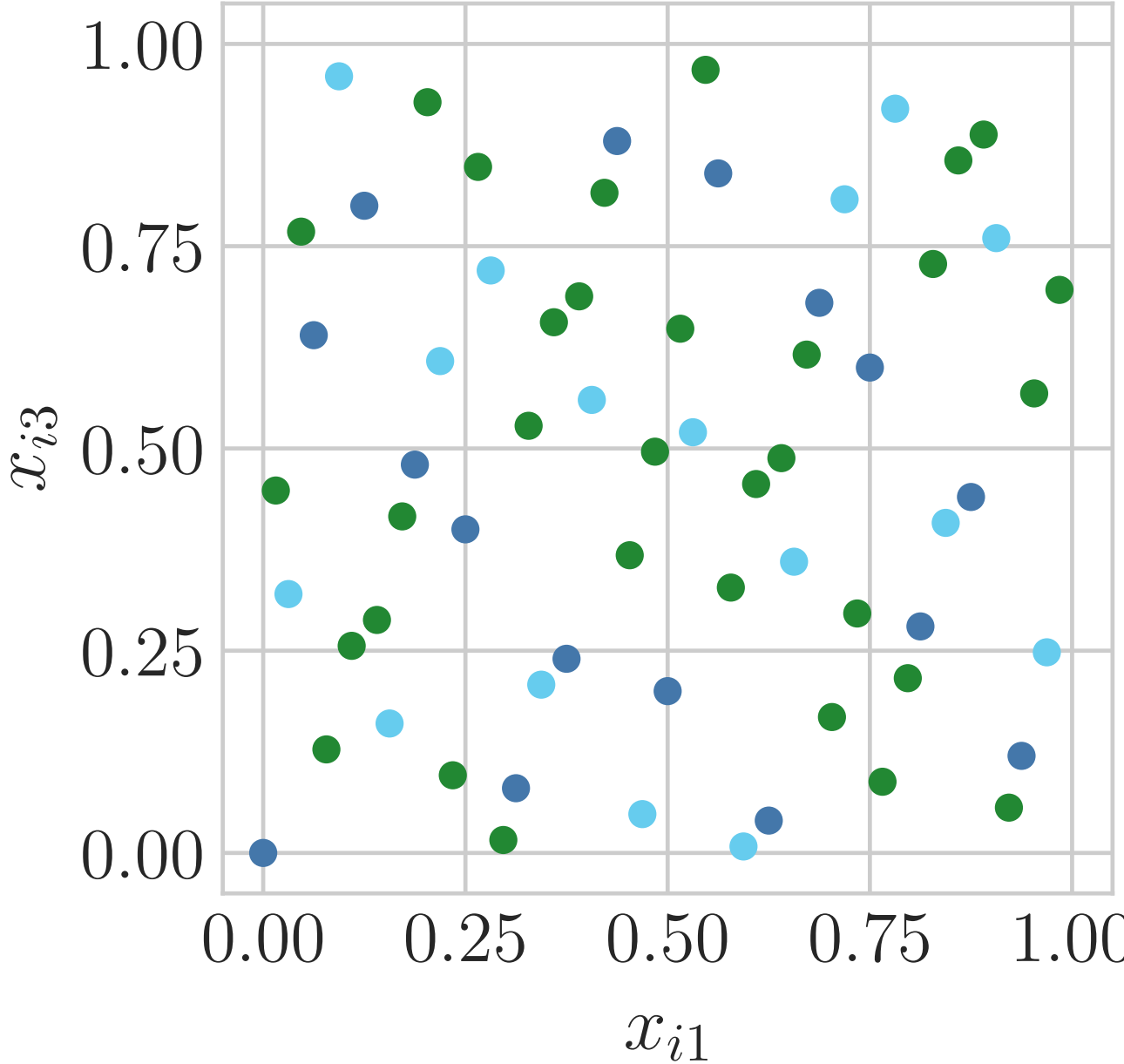
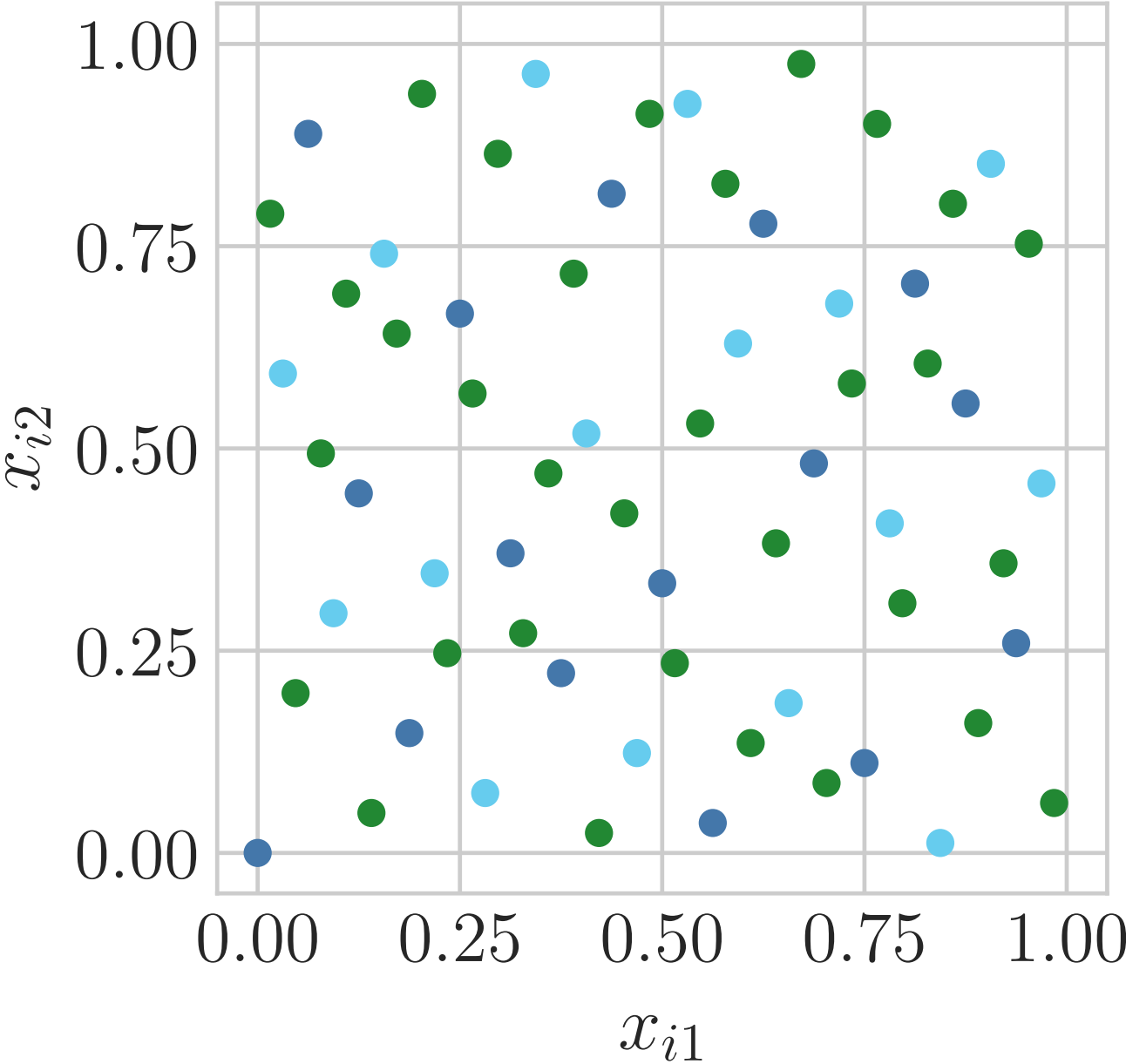


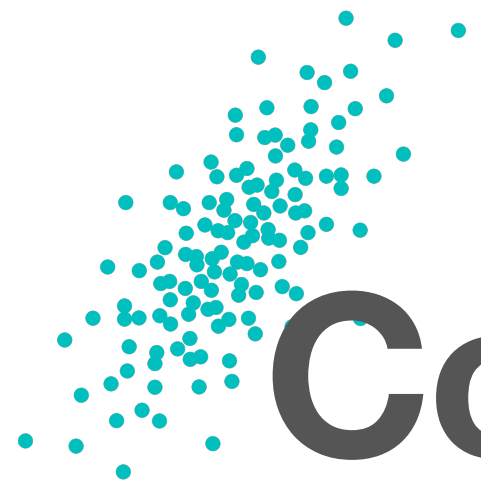
## Sobol' LD





Halton LD





# Component-by-component search

The generators for digital nets may be chosen by number theory, but in most cases the generators are chosen by computer search to minimize the discrepancy.

The component-by-component search method

1. Choose a generator for a 1-D sequence that minimizes the discrepancy over a range of  $n$ , set  $\ell = 1$
2. Freeze the first  $\ell$  dimensions of the generator
3. Choose the  $\ell + 1$  component of the generator by minimizing the discrepancy
4. Increases  $\ell$  by one
5. If  $\ell < d$ , go to step 2; otherwise you are done



# Randomizing low discrepancy sequences

Want to make  $\mathbf{x}_{\text{rand},i} \sim \mathcal{U}[0,1]^d$ , while  $\{\mathbf{x}_{\text{rand},i}\}_{i=0}^{n-1}$  is low discrepancy

- Random **shifts**:  $\mathbf{x}_{\text{rand},i} = \mathbf{x}_i + \Delta \bmod \mathbf{1}$ ,  $\Delta \sim \mathcal{U}[0,1]^d$ , especially for lattices and Kronecker
- Random **digital shifts**:  $\mathbf{x}_{\text{rand},i} = \mathbf{x}_i \oplus \Delta \bmod \mathbf{1}$ ,  $\Delta \sim \mathcal{U}[0,1]^d$ , especially for digital sequences
- **Linear matrix scrambling** or **nested uniform scrambling**, especially for digital sequences





# Benefits of randomizing

- Unbiased estimator,  $\hat{\mu} = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_{\text{rand},i})$ , of  $\mathbb{E}[f(X)]$  for  $X \sim \mathcal{U}[0,1]^d$
- $\mathbf{x}_{\text{rand},i}$  almost surely never on the boundary of  $\mathcal{U}[0,1]^d$ , so transformations to the Gaussian distribution will be finite
- Root mean squared error for digital sequences with linear matrix or nested uniform scrambling can be nearly  $\mathcal{O}(n^{-3/2})$